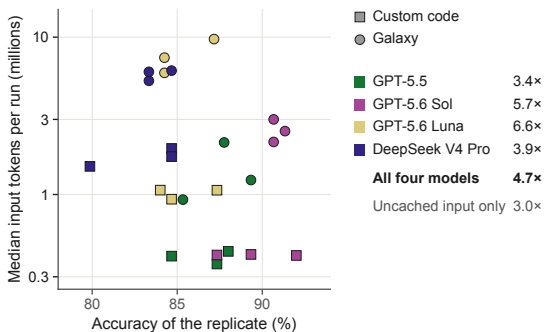


a The trade: same accuracy, more input tokens

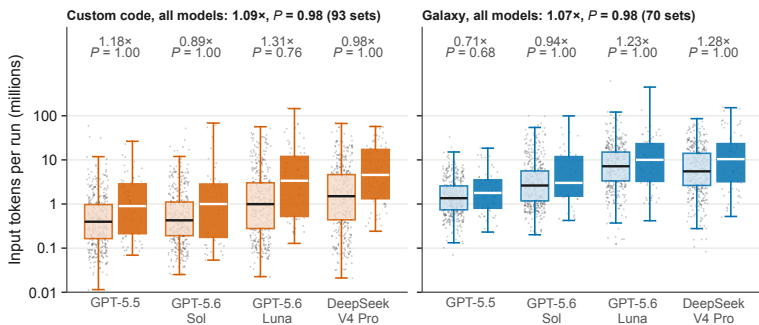
BigBench-Verified-50 and CompBioBench; each point is one replicate (one run per task)
Numbers: times more input tokens in Galaxy on the same task



b Incorrect runs cost no more than correct runs of the same task

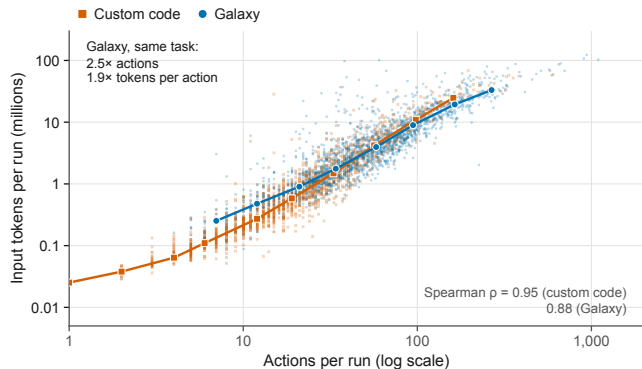
Numbers: input tokens of incorrect runs relative to correct runs of the same task and model

Custom code, correct Custom code, incorrect Galaxy, correct Galaxy, incorrect



c Cost follows interaction

Correct runs; lines join median input tokens in bins of actions



d Finding tools is half of what Galaxy sends back

Each Galaxy interaction is a request from the agent and a text reply from Galaxy; the model reads every reply and rereads it at each later step (all Galaxy runs)

